DANG: Prototype-Anchored Foundation Features for Label-Efficient and Robust Polyp Segmentation

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October 18, 2026

Abstract

Polyp segmentation networks that reach $\sim$ 93% mean Dice (mDice) in domain still fail where it matters clinically: when only a few frames are labelled, on tiny or flat low-contrast lesions, and under acquisition shifts such as narrow-band imaging (NBI), low light, glare and blur. We present **DANG**, a dual-stream network that pairs a trainable ConvNeXt-Tiny with a frozen DINOv2-Base and uses the foundation features through three components: a dual-prototype margin head on multi-depth DINOv2 tokens, a prototype-anchored spatial support envelope (PASSE) that suppresses far-field false positives, and an uncertainty-guided directional strip decoder. We train on Kvasir-SEG only and test on deduplicated external cohorts and two diagnostic stress slices over five paired seeds, retraining PraNet, Polyp-PVT and EMCAD under the same protocol. With 40 labelled images DANG reaches 51.0% mDice on tiny/camouflaged lesions and 55.8% on synthetic adverse endoscopy, ahead of every retrained model by 7–27 pp, and its Boundary F1 exceeds theirs by 5–34 pp. A control that adds the same frozen DINOv2 by plain concatenation gains nothing in this regime, so the improvement comes from how the foundation features are used, not from the extra parameters. With 800 labelled images DANG beats PraNet but is statistically tied with Polyp-PVT and EMCAD. On the standard PraNet training split it matches a faithful Polyp-PVT reproduction on four test sets and exceeds it on ETIS-LaribPolypDB (84.4% vs. 78.9%). These gains cost efficiency: DANG has 130M parameters (43M trainable) and runs at 17 FPS on a T4, against 25–31M parameters and 37–42 FPS for the retrained models.

1 Introduction

Colorectal cancer remains a leading cause of cancer death, and detecting and resecting polyps during colonoscopy reduces its incidence (Jha et al., 2020; Ali et al., 2023). Pixel-level polyp segmentation supports this by delineating lesion margins. Its benchmarks, however, are close to saturation in domain: a ConvNeXt-Tiny (Liu et al., 2022) with a Feature Pyramid Network (Lin et al., 2017) already reaches about 93% mDice on Kvasir-SEG (Jha et al., 2020), and architectural changes move this number by less than one point. In-domain scores therefore say little about whether a model is more robust.

Deployment stresses a model along three axes that in-domain benchmarks do not exercise:

1. **Few labels (S_1).** Pixel-accurate contours require expert time, so a new centre may label only tens of frames. Trained on 40 images, CNNs overfit to source textures and their boundaries degrade on external centres.

2. **Tiny and camouflaged lesions (S_2).** Flat or diminutive polyps covering under 1.5% of the frame, or with low contrast to the surrounding mucosa, are fragmented or missed by stride-16/32 features, especially on ETIS-LaribPolypDB (Silva et al., 2014) and CVC-ColonDB (Tajbakhsh et al., 2016).

3. **Acquisition shift (S_3).** Procedures switch between white-light and NBI-like spectra and suffer illumination fall-off, sensor noise, specular glare and motion blur.

Vision foundation models such as DINOv2 (Oquab et al., 2024) and SAM (Kirillov et al., 2023) provide features that transfer across domains, but their 14-pixel patch grid is too coarse for lesion boundaries. Concatenating such features with a CNN is the obvious remedy; we show that it does not help when labels are scarce.

We present **DANG** (Figure 1) and make four contributions:

- **A deduplicated multi-centre stress benchmark.** Perceptual-hash and MD5 deduplication across Kvasir-SEG, six-centre PolypGen (Ali et al., 2023) and the four PraNet external sets (Fan et al., 2020), with a 40-image few-shot track and two stress slices: natural tiny/camouflaged lesions and synthetic adverse acquisitions (Figure 2).

- **Prototype-anchored use of foundation features.** Multi-depth DINOv2 tokens feed a dual-prototype margin head with test-time self-support, and the **prototype-anchored spatial support envelope (PASSE)** turns this margin into a mask that suppresses false positives far from the lesion and reinforces low-contrast lesion cores.

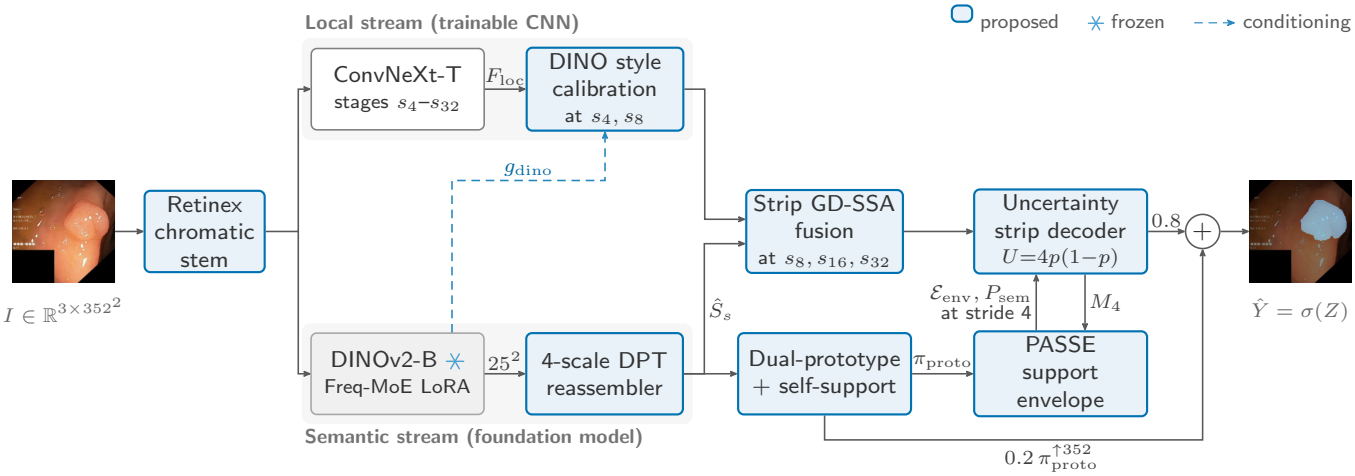


Figure 1: **DANG fuses a trainable CNN stream with a frozen foundation-model stream and gates the decoder with a prototype-derived support envelope.** A 352×352 frame passes through a Retinex chromatic stem and splits into a ConvNeXt-Tiny local stream and a frozen DINOv2-Base semantic stream with low-rank adapters on its last four blocks. Tokens from four depths are reassembled by a 4-scale DPT; the global DINOv2 vector g_{dino} (dashed) conditions style calibration at strides 4 and 8. Strip fusion aligns the streams at strides 8–32, the dual-prototype head produces the margin map π_{proto}, and PASSE combines it with the coarsest decoder logit M_4 into a support envelope that modulates the stride-4 decoder logit; the final logit adds $0.2\,\pi_{\text{proto}}$ (Eq. 5). Blue boxes are the proposed components. The thumbnails are a Kvasir test frame and its ground-truth mask, shown for illustration only.

- **An uncertainty-guided strip decoder.** Fused horizontal, vertical and dilated depthwise strips, gated by the Bernoulli boundary variance $4p(1-p)$, refine boundaries where the coarser prediction is ambiguous.
- **A controlled five-seed evaluation against retrained state of the art.** PraNet (Fan et al., 2020), Polyp-PVT (Dong et al., 2021) and EMCAD (Rahman et al., 2024) are retrained under our protocol, a concatenation control isolates the effect of the foundation features, and a standard-split experiment ties our protocol to published numbers. We report where DANG does not win, including the full-data regime and inference cost, and that its Retinex stem does not contribute.

2 Related Work

Polyp segmentation. Architectures progressed from U-Net (Ronneberger et al., 2015) and UNet++ (Zhou et al., 2018) to reverse-attention networks: PraNet (Fan et al., 2020) mines boundary cues with parallel partial decoders and reverse attention, and CaraNet (Lou et al., 2022) adds axial context for small objects. Transformer encoders improved multi-centre generalisation in Polyp-PVT (Dong et al., 2021) and TransFuse (Zhang et al., 2021), and EMCAD (Rahman et al., 2024) pairs a PVTv2 encoder with an efficient multi-scale convolutional attention decoder.

Foundation models for polyps. SAM-Mamba (Dutta et al., 2024) and SAM-MaGuP (Dutta et al., 2025) adapt SAM with Mamba-based priors and report strong zero-shot generalisation, at about 103M parameters and 423 GFLOPs (Agnihotri et al., 2026a). A second line distils foundation models into small networks: Polyp-DiFoM (Agnihotri et al., 2025) distils SAM and DINOv2 into lightweight baselines, LiteBounD (Agnihotri et al., 2026a) adds boundary-guided frequency-aware distillation, and Lite-π (Agnihotri et al., 2026b) aligns foundation-model-specific prototypes with a lightweight network for real-time inference. DANG keeps the foundation model at inference and studies how to use its features when labels are scarce; it is complementary to distillation, which could compress it.

Adapting foundation models and directional operators. Parameter-efficient fine-tuning such as LoRA (Hu et al., 2022) and dense-prediction heads such as DPT (Ranftl et al., 2021) adapt frozen ViTs to segmentation. State-space models (Mamba (Gu and Dao, 2023), VMamba (Liu et al., 2024), U-Mamba (Ma et al., 2024)) capture long-range context with directional scans; our strip decoder approximates directional context with fused depthwise strip convolutions instead.

3 Method

DANG maps an RGB frame $I \in \mathbb{R}^{3\times 352\times 352}$ to a polyp probability map $\hat{Y} = \sigma(Z)$ (Figure 1).

3.1 Two streams

The local stream is ConvNeXt-Tiny, producing features $F_{\text{loc}}^{(s)}$ at strides $s \in \{4, 8, 16, 32\}$. The semantic stream is DINOv2-Base, frozen, with rank-4 convolutional LoRA adapters on its last four blocks that route low- and high-frequency token components to experts with dilations 1–3. A Retinex chromatic stem (Land and McCann, 1971) removes specular highlights and divides out a smooth illumination estimate before both streams, and the global DINOv2 vector conditions a feature-space style calibration (with MixStyle (Zhou et al., 2021)) at strides 4 and 8. The stem is trained without synthetic corruptions; Section 5.3 shows that it does not contribute.

3.2 Multi-scale reassembly and dual-prototype head

We take DINOv2 tokens from blocks $\{2, 5, 8, 11\}$ on a 25×25 grid and project each to $D = 192$ channels, S_ℓ. A learned depth-mixing matrix $W = \text{Softmax}(\Theta) \in \mathbb{R}^{4\times4}$ produces one semantic map per decoder scale,

$$\tilde{S}_s = \sum_\ell W_{s,\ell} S_\ell, \qquad \hat{S}_s = \phi_s\big(\text{Up}_s(\tilde{S}_s, F_{\text{loc}}^{(s)})\big), \tag{1}$$

where Up_s at strides 8 and 16 predicts 3×3 reassembly kernels from the CNN features, so that upsampled semantic features snap to CNN edges.

The prototype head projects the reassembled tokens to L_2-normalised embeddings $\hat{Z}_{u,v} \in \mathbb{R}^{128}$ and compares them with $K = 8$ foreground prototypes p_k^+ and $K = 8$ background prototypes p_k^-, plus foreground and background anchors $a^\pm$ that track ground-truth token means during training:

$$\begin{aligned}\pi_{\text{prior}}(u,v) = \tfrac{\tau}{\beta}\Big(&\log \textstyle\sum_k e^{\beta\langle \hat{Z}_{u,v}, p_k^+\rangle} \\ &- \log \textstyle\sum_k e^{\beta\langle \hat{Z}_{u,v}, p_k^-\rangle}\Big) \\ &+ w_{\text{a}}\big(\langle \hat{Z}_{u,v}, a^+\rangle - \langle \hat{Z}_{u,v}, a^-\rangle\big) + b.\end{aligned} \tag{2}$$

At test time no mask is available, so the head also computes image-specific self-support centroids $c^+ = \text{Norm}(\sum p_{u,v}^2 \hat{Z}_{u,v})$ and $c^- = \text{Norm}(\sum (1-p_{u,v})^2 \hat{Z}_{u,v})$ with $p = \sigma(\pi_{\text{prior}})$, giving the margin map

$$\pi_{\text{proto}} = \pi_{\text{prior}} + w_{\text{ss}}\big(\langle \hat{Z}, c^+\rangle - \langle \hat{Z}, c^-\rangle\big). \tag{3}$$

3.3 Prototype-anchored spatial support envelope (PASSE)

CNN texture far from the lesion can cross the decision threshold and produce isolated false positives. PASSE builds a coarse semantic probability $P_{\text{sem}} = \sigma(0.5\, M_4^{\uparrow 25} + 0.5\, \pi_{\text{proto}})$ from the coarsest decoder logit M_4 and the prototype margin, dilates it into a smooth envelope

$$\mathcal{E} = \text{AvgPool}_{3\times3}\big(\text{MaxPool}_{5\times5}(P_{\text{sem}})\big), \tag{4}$$

and uses it to modulate the stride-4 decoder logit Z_{88}:

$$\begin{aligned}Z_{88}^{\text{P}} = Z_{88} &- \lambda_{\text{out}}\,\text{ReLU}\big(t_{\text{out}} - \mathcal{E}^{\uparrow 88}\big) \\ &+ \lambda_{\text{in}}\,\text{ReLU}\big(P_{\text{sem}}^{\uparrow 88} - t_{\text{in}}\big), \\ Z = 0.8\,\big(Z_{88}^{\text{P}}\big)^{\uparrow 352} &+ 0.2\,\pi_{\text{proto}}^{\uparrow 352}.\end{aligned} \tag{5}$$

The first term suppresses logits outside the envelope and the second reinforces confident lesion cores ($t_{\text{out}} = 0.22$, $\lambda_{\text{out}} = 10$, $t_{\text{in}} = 0.38$, $\lambda_{\text{in}} = 1.2$).

3.4 Strip fusion and uncertainty-guided decoder

At strides 8–32 the two streams are fused with a gated directional strip operator,

$$\begin{aligned}\text{Strip}(U, C) = U &+ g_h \odot \mathcal{K}_{7\times1}\mathcal{K}_{1\times11}(U) \\ &+ g_v \odot \mathcal{K}_{1\times7}\mathcal{K}_{11\times1}(U) + g_d \odot \mathcal{K}_{3\times3,d=3}(U),\end{aligned} \tag{6}$$

where $\mathcal{K}$ are depthwise convolutions and the gates $(g_h, g_v, g_d) = \sigma(\text{Conv}_{1\times1}([U, C]))$ are conditioned on the other stream. Each top-down decoder stage receives the upsampled logit of the coarser stage, converts it to a foreground probability A, and computes the boundary uncertainty $U = 4A(1 - A)$, which peaks at $A = 0.5$, and an outer ring $\text{MaxPool}_{5\times5}(A) - A$. Both maps gate the strip operator and a Sobel edge refinement, so computation concentrates on ambiguous borders.

3.5 Training objective

Every model in this paper, including the baselines, is trained with the same loss:

$$\begin{aligned}\mathcal{L} = &\mathcal{L}_{\text{BCE+Dice}}^{w}(Z, Y) + 0.25\,\mathcal{L}_{\text{Tversky}}^{w}(Z, Y) \\ &+ 0.1\,\mathcal{L}_{\text{far}}(Z, Y) + \tfrac{0.25}{3}\textstyle\sum_{j=2}^{4}\mathcal{L}_{\text{BCE+Dice}}^{w}(M_j, Y) \\ &+ 0.35\,\mathcal{L}_{\text{proto}}(\pi_{\text{proto}}, Y_{25}).\end{aligned} \tag{7}$$

The per-image weights w grow as lesion area shrinks, the Tversky trade-off shifts towards recall for small lesions, $\mathcal{L}_{\text{far}}$ penalises predicted probability outside dilations of the ground truth, and M_j are deep-supervision logits. $\mathcal{L}_{\text{proto}}$ applies the same Dice, cross-entropy and Tversky terms to the prototype margin and is present only in models with the prototype head. The published baselines are trained with their official losses; one Polyp-PVT run uses Eq. (7) as a loss-matched control.

4 Experimental Setup

4.1 Data and cohorts

We deduplicate every evaluation set against the training data with MD5 hashes and a 64-bit perceptual difference hash (dHash ≤ 4), and remove Kvasir frames that public composites redistribute under other names.

- **Kvasir-SEG** (Jha et al., 2020): 800 training images (*Full* track), a fixed 40-image subset of them (5%, the few-shot track S_1), 95 validation and 90 in-domain test images.
- **PolypGen** (Ali et al., 2023): six centres; a development set ($n = 258$) and an extended holdout ($n = 974$) that was not inspected during design and serves as a confirmation cohort.
- **PraNet external sets** (Fan et al., 2020): CVC-300 (60), CVC-ClinicDB (61), CVC-ColonDB (379) and ETIS-LaribPolypDB (196), 696 images after deduplication. ClinicDB is unseen because we train on Kvasir only.
- S_2**, tiny and camouflaged** ($n = 310$): natural lesions from PolypGen-Dev, ETIS and ColonDB covering under 1.5% of the frame, or under 3% with low boundary contrast.
- S_3**, adverse acquisition** ($n = 258$): PolypGen-Dev frames under one of four deterministic corruptions: NBI-like spectral shift, low light with vignetting and shot noise, specular glare with blur, and a combined severe mode.

S_2 and S_3 are diagnostic slices of the external cohorts, not independent patient cohorts, and S_3 is synthetic.

4.2 Training and evaluation

All models are trained at 352×352 with batch size 16, AdamW, warm-up followed by cosine decay, an exponential moving average of the weights, and mixed precision. The few-shot track trains for 40 epochs and the Full track for 20. The checkpoint is selected by Kvasir validation mDice only. Every model, including the published baselines, uses the same data, augmentation, schedule, seeds and evaluation code; for a given seed all models see the same 40-image subset.

Predictions are thresholded at 0.5. We report mDice, Boundary F1 (BF1, 2-pixel tolerance) and the 95th-percentile Hausdorff distance (HD95, pixels at 352×352). Differences between models are tested with paired t-tests over seeds 42–46, and each family of 12 comparisons is corrected with Holm's procedure. Inference speed is measured at batch size 1 in FP16 on an NVIDIA T4.

5 Results

5.1 Comparison with retrained state of the art

Table 1 compares DANG with PraNet, Polyp-PVT and EMCAD retrained under our protocol.

Few labels. With 40 labelled images DANG has the highest mDice and BF1 on every few-shot cohort. Against Polyp-PVT, the strongest retrained model, it gains 8.2 pp mDice on tiny/camouflaged lesions, 16.6 pp on adverse acquisitions and 7.6 pp on the PraNet external sets, and 15.8–20.8 pp BF1. All BF1 and adverse-acquisition differences survive Holm correction; the tiny/camouflaged and PraNet-external mDice differences do not (adjusted $p = 0.085$). EMCAD, whose decoder is designed for efficiency, degrades most in this regime. Training Polyp-PVT with our loss does not close the gap and makes it unstable across seeds (SD up to 12.7 pp), so the gap is not a loss artefact.

Full data. With 800 labelled images DANG still beats PraNet on all cohorts, but its leads over Polyp-PVT (+1.4 pp on ETIS, +0.7 pp on PraNet-Ext, −0.1 pp on PolypGen-Ext) and EMCAD are not significant. DANG's advantage is therefore specific to label-scarce and shifted conditions.

Qualitative results. Figure 3 shows predicted masks on randomly drawn stress frames. The tiny ColonDB lesion is missed by every model. On the camouflaged, low-light and combined-corruption frames DANG recovers the lesion with Dice 82, 53 and 60, and on the low-light frame it is the only model that finds the lesion at all.

5.2 Standard split and published numbers

Our protocol trains on Kvasir only, so its numbers are not comparable to the literature, which trains on 900 Kvasir and 550 ClinicDB images. To tie the two, we trained DANG and Polyp-PVT on this standard split for 50 epochs, kept the final weight average without test-set checkpoint selection, and evaluated at native resolution (three seeds; Table 2). Our Polyp-PVT reproduction is within 1.1 pp of its published scores on every set, which validates the pipeline. DANG matches it on Kvasir, ClinicDB, ColonDB and CVC-300 (all differences within ±1.1 pp, not significant) and exceeds it on ETIS-LaribPolypDB by 5.5 pp mDice ($p = 0.026$) and 11.1 pp BF1 ($p = 0.007$; paired, $n = 3$, uncorrected). ETIS, the set with the smallest and least textured lesions, is where the few-shot results predict an advantage.

5.3 Where the gain comes from

Table 3 and Figures 4–5 isolate the components.

Foundation features alone do not help. The concatenation control (S) has the same frozen DINOv2, the same adapters and nearly the same size as DANG (120M parameters), yet in the few-shot regime it is indistinguishable from the 31M-parameter baseline (44.9% vs. 44.7% on $S_1 \times S_2$, 46.3% vs. 45.6% on $S_1 \times S_3$). DANG, with 10M more parameters, gains 6.1 pp ($p = 0.003$) and 9.5 pp ($p < 0.001$) over the control on these cohorts and 14.1 pp

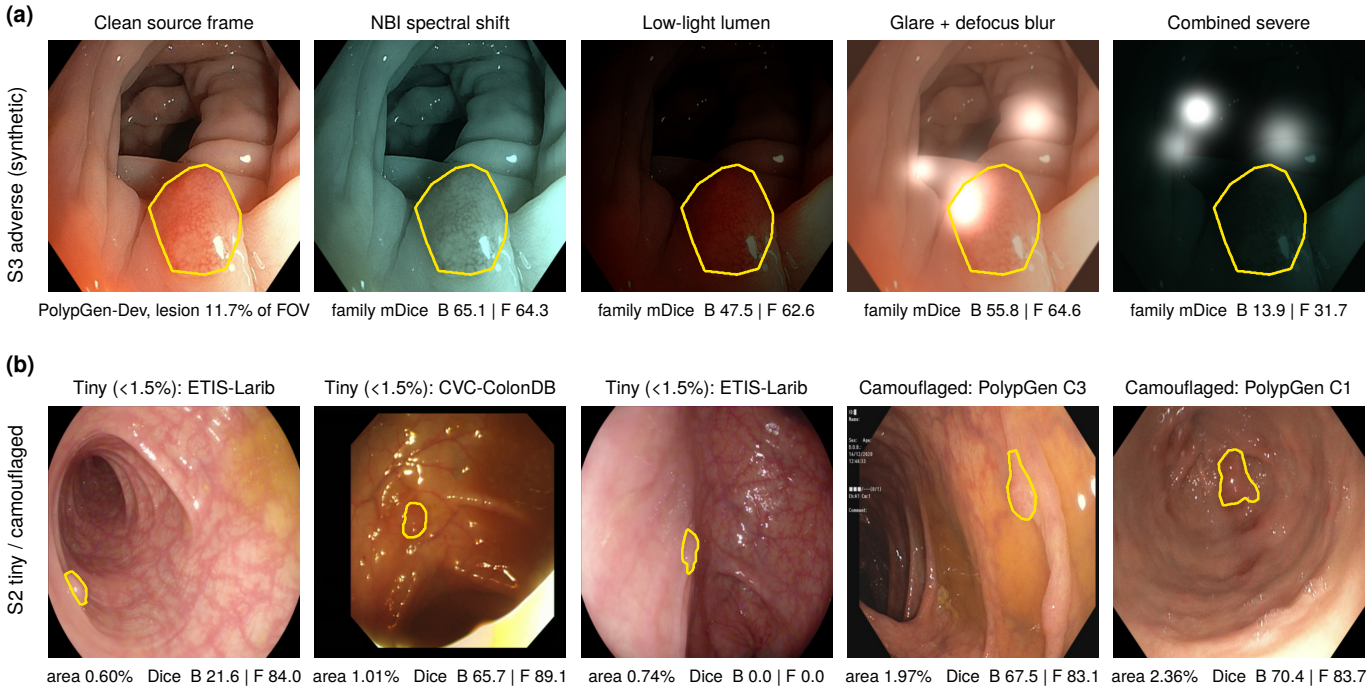


Figure 2: **The stress cohorts contain frames on which a few-shot CNN fails outright. (a)** One PolypGen-Dev frame shown clean and under the four S_3 corruptions; in the benchmark each frame receives one of the four. Below each corruption: cohort-level mDice of the ConvNeXt-T baseline (B) and DANG (F) on that corruption family, few-shot track, mean of 5 seeds. **(b)** Five S_2 lesions drawn at random (three tiny, two camouflaged). Below each: lesion area and per-image Dice of B and F, few-shot track, mean of 5 seeds. Yellow contours are ground truth; predicted masks are shown in Figure 3.

Table 1: **DANG leads every retrained model when labels are scarce and ties the strongest ones with full data.** All models trained on Kvasir-SEG with the same data, augmentation, schedule and evaluation; mean ± SD over 5 seeds. Few-shot columns use 40 labelled images, Full columns 800. *: worse than DANG after Holm correction (paired t-test, $n = 5$). Params are total parameters in millions (DANG: 43.3M trainable, 86.6M frozen); FPS at batch size 1, FP16, NVIDIA T4. "Polyp-PVT, our loss" is trained with Eq. (7) instead of its official loss.

			$S_1 \times S_2$ **tiny/camo**		$S_1 \times S_3$ **adverse**		$S_1 \rightarrow$ **PraNet-Ext**		**Full, mDice**		
Model	Params	FPS	mDice	BF1	mDice	BF1	mDice	BF1	ETIS	PraNet-Ext	PGen-Ext
ConvNeXt-T + FPN	31.0	91.9	$44.7_{\pm 3.8}$	$16.2_{\pm 3.9}$*	$45.6_{\pm 4.0}$*	$15.2_{\pm 2.0}$*	$66.6_{\pm 3.8}$	$28.4_{\pm 4.8}$*	$72.5_{\pm 2.7}$*	$78.7_{\pm 1.5}$*	$78.7_{\pm 0.8}$*
PraNet (Fan et al., 2020)	30.5	40.6	$40.4_{\pm 1.5}$*	$21.6_{\pm 1.5}$*	$48.5_{\pm 3.5}$*	$22.4_{\pm 1.6}$*	$62.9_{\pm 1.7}$*	$32.2_{\pm 1.2}$*	$69.7_{\pm 2.0}$*	$76.7_{\pm 1.1}$*	$76.9_{\pm 0.9}$*
Polyp-PVT (Dong et al., 2021)	25.1	42.4	$42.9_{\pm 4.0}$	$14.8_{\pm 4.2}$*	$39.3_{\pm 4.5}$*	$10.7_{\pm 2.6}$*	$64.3_{\pm 4.5}$	$21.5_{\pm 6.5}$*	$77.5_{\pm 2.2}$	$81.1_{\pm 0.8}$	$80.2_{\pm 1.6}$
Polyp-PVT, our loss	25.1	43.0	$43.9_{\pm 10.2}$	$16.4_{\pm 6.8}$	$38.3_{\pm 11.4}$	$12.1_{\pm 4.9}$	$62.2_{\pm 12.7}$	$22.7_{\pm 9.2}$	$77.2_{\pm 1.0}$	$80.4_{\pm 0.8}$	$79.4_{\pm 1.0}$
EMCAD (Rahman et al., 2024)	26.8	36.5	$29.6_{\pm 3.1}$*	$6.0_{\pm 2.5}$*	$29.2_{\pm 3.5}$*	$3.3_{\pm 1.1}$*	$49.7_{\pm 4.9}$*	$8.0_{\pm 4.1}$*	$77.5_{\pm 2.1}$	$81.0_{\pm 0.9}$	$79.7_{\pm 0.9}$
DANG	129.9	17.4	$\mathbf{51.0}_{\pm 1.7}$	$\mathbf{30.7}_{\pm 1.2}$	$\mathbf{55.8}_{\pm 1.2}$	$\mathbf{27.2}_{\pm 1.7}$	$\mathbf{71.9}_{\pm 0.3}$	$\mathbf{42.2}_{\pm 1.6}$	$\mathbf{78.9}_{\pm 1.5}$	$\mathbf{81.8}_{\pm 0.6}$	$\mathbf{80.1}_{\pm 0.4}$

Table 2: **Standard PraNet training split (1,450 images), mDice (%) at native resolution.** Mean ± SD over 3 seeds for our runs; published Polyp-PVT scores from Dong et al. (2021).

Model	Kvasir	ClinicDB	ColonDB	ETIS	CVC-300
Polyp-PVT (published)	91.7	93.7	80.8	78.7	90.0
Polyp-PVT (ours)	$\mathbf{92.2}_{\pm 0.6}$	$92.6_{\pm 0.3}$	$\mathbf{80.6}_{\pm 0.7}$	$78.9_{\pm 1.2}$	$89.9_{\pm 0.2}$
DANG	$91.9_{\pm 0.3}$	$\mathbf{93.7}_{\pm 0.7}$	$80.2_{\pm 0.9}$	$\mathbf{84.4}_{\pm 0.4}$	$\mathbf{90.8}_{\pm 1.1}$

BF1 on tiny/camouflaged lesions. Parameter count therefore does not explain DANG's advantage; the prototype, PASSE and strip-decoder path does.

Gains over the baseline. DANG raises few-shot BF1 by 11.9–14.4 pp on all three cohorts and mDice on adverse acquisitions by 10.2 pp, all significant after correction. With full data it gains 6.3 pp on ETIS and 3.2 pp on the PraNet external sets while in-domain accuracy is unchanged (92.95% vs. 92.92% on Kvasir test). On the untouched PolypGen extended holdout the gain is smaller but significant (+1.4 pp, $p = 0.009$; +1.2 pp on the subset that shares no video sequence with the development set,

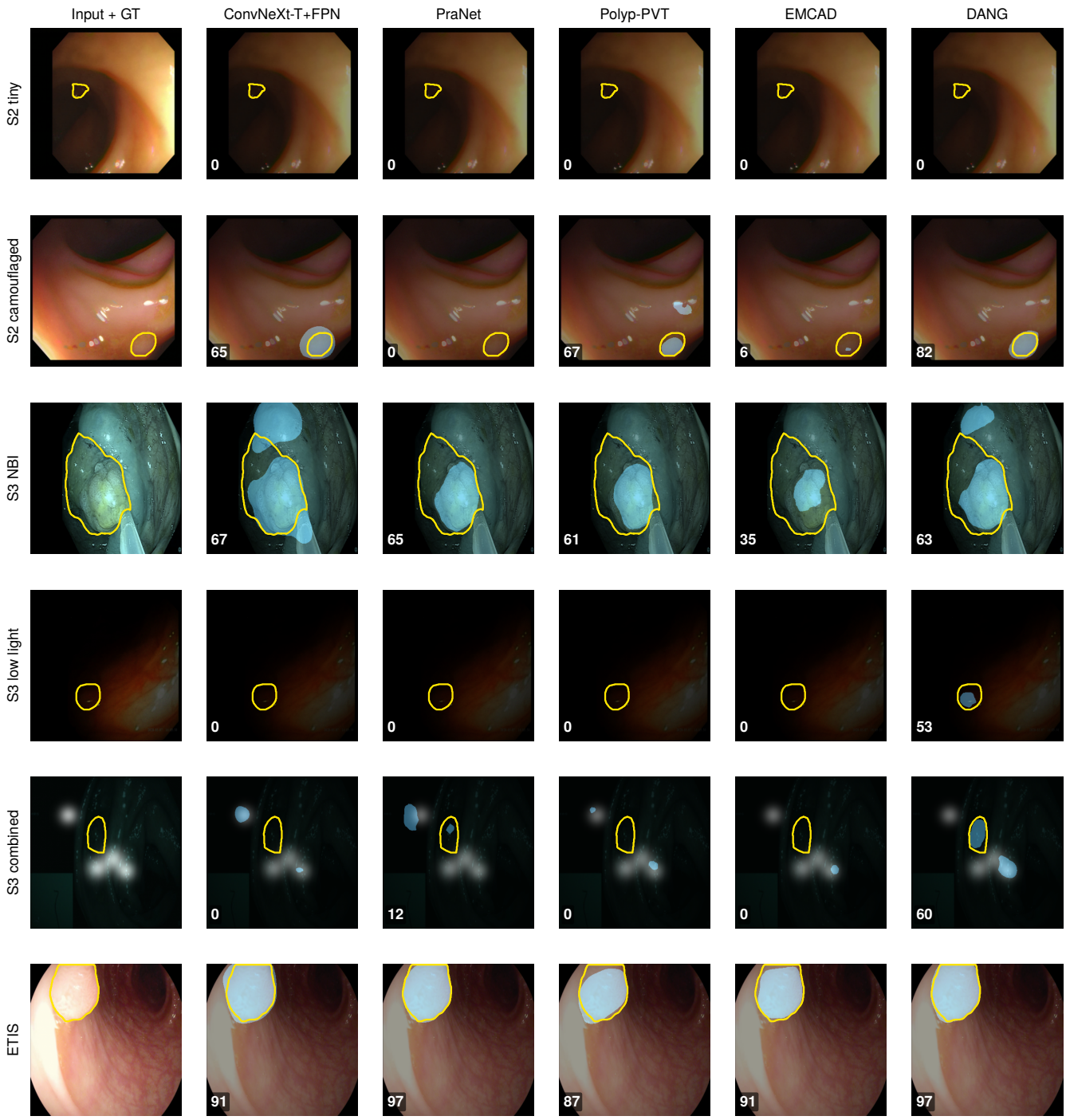


Figure 3: **Predicted masks on stress frames, few-shot track.** Each row is one frame drawn at random (not curated) from the named stratum; columns are the ConvNeXt-T baseline, PraNet, Polyp-PVT, EMCAD and DANG, each trained on the same 40 Kvasir images (seed 42). Blue fill: prediction at threshold 0.5; yellow contour: ground truth; number: per-image Dice.

$p = 0.004$). Full-data S_3 mDice rises from 55.8% to 67.4%, most on low-light (+18.7 pp) and combined-severe frames (+23.0 pp).

The stem is a negative result. Removing the Retinex stem and style calibration (C) keeps almost all of the improvement, and no mDice difference between C and F survives correction. The only uncorrected mDice effect of the stem is a loss of 4.3 pp on NBI frames ($p = 0.008$). One plausible reason, which we did not test, is that the stem standardises colour towards white-light Kvasir statistics and so maps the green-dominant NBI palette onto a

Table 3: **Ablation over five paired seeds (mean ± SD).** (B) ConvNeXt-T + FPN; (S) dual-stream control with the same frozen DINOv2 and adapters, fused by concatenation into an FPN; (C) DANG without the Retinex stem and style calibration; (F) full DANG. Difference rows give the paired mean difference and the p-value of a paired t-test ($n = 5$); **bold** differences remain significant after Holm correction over the 12 columns. HD95 in pixels; PGen-Ext is the PolypGen extended holdout.

	$S_1 \times S_2$ **tiny/camo** (310)			$S_1 \times S_3$ **adverse** (258)			$S_1 \rightarrow$ **PraNet-Ext** (696)			**Full** ($n = 800$), **mDice**		
Model	mDice↑	BF1↑	HD95↓	mDice↑	BF1↑	HD95↓	mDice↑	BF1↑	HD95↓	ETIS↑	PraNet-Ext↑	PGen-Ext↑
(B) Baseline	$44.74_{\pm 3.79}$	$16.22_{\pm 3.87}$	$72.76_{\pm 8.85}$	$45.59_{\pm 4.02}$	$15.23_{\pm 1.97}$	$87.18_{\pm 1.61}$	$66.59_{\pm 3.84}$	$28.45_{\pm 4.82}$	$58.76_{\pm 14.41}$	$72.55_{\pm 2.68}$	$78.66_{\pm 1.55}$	$78.71_{\pm 0.76}$
(S) Concatenation	$44.92_{\pm 0.82}$	$16.59_{\pm 2.50}$	$75.55_{\pm 5.48}$	$46.33_{\pm 1.85}$	$16.40_{\pm 1.61}$	$88.21_{\pm 5.20}$	$67.23_{\pm 1.15}$	$29.20_{\pm 4.19}$	$63.27_{\pm 9.45}$	$72.95_{\pm 3.85}$	$79.73_{\pm 1.04}$	$79.35_{\pm 1.43}$
(C) No stem	$50.06_{\pm 1.36}$	$27.37_{\pm 1.64}$	$65.76_{\pm 3.04}$	$\mathbf{57.24}_{\pm 2.03}$	$25.64_{\pm 1.10}$	$\mathbf{71.42}_{\pm 2.40}$	$71.55_{\pm 0.76}$	$40.36_{\pm 1.05}$	$\mathbf{49.34}_{\pm 1.12}$	$77.49_{\pm 0.66}$	$\mathbf{81.93}_{\pm 0.77}$	$79.74_{\pm 0.71}$
(F) **DANG**	$\mathbf{51.05}_{\pm 1.71}$	$\mathbf{30.65}_{\pm 1.16}$	$\mathbf{64.68}_{\pm 1.76}$	$55.82_{\pm 1.17}$	$\mathbf{27.17}_{\pm 1.72}$	$78.74_{\pm 2.70}$	$\mathbf{71.86}_{\pm 0.34}$	$\mathbf{42.25}_{\pm 1.58}$	$49.55_{\pm 2.73}$	$\mathbf{78.88}_{\pm 1.50}$	$81.82_{\pm 0.64}$	$\mathbf{80.09}_{\pm 0.45}$
F − B	+6.31 (.014)	**+14.44** (.002)	−8.08 (.135)	**+10.23** (.005)	**+11.95** (.002)	**−8.44** (.002)	+5.27 (.038)	**+13.80** (.007)	−9.21 (.258)	**+6.33** (.005)	**+3.16** (.003)	**+1.38** (.009)
C − B	+5.32 (.017)	**+11.15** (.001)	−7.01 (.187)	**+11.65** (.003)	**+10.42** (<.001)	**−15.76** (<.001)	+4.95 (.037)	**+11.91** (.003)	−9.42 (.207)	+4.94 (.019)	+3.28 (.019)	+1.03 (.127)
F − C	+0.99 (.218)	+3.29 (.022)	−1.08 (.551)	−1.42 (.112)	+1.53 (.225)	+7.32 (.012)	+0.32 (.489)	+1.89 (.152)	+0.20 (.897)	+1.39 (.106)	−0.11 (.847)	+0.35 (.471)

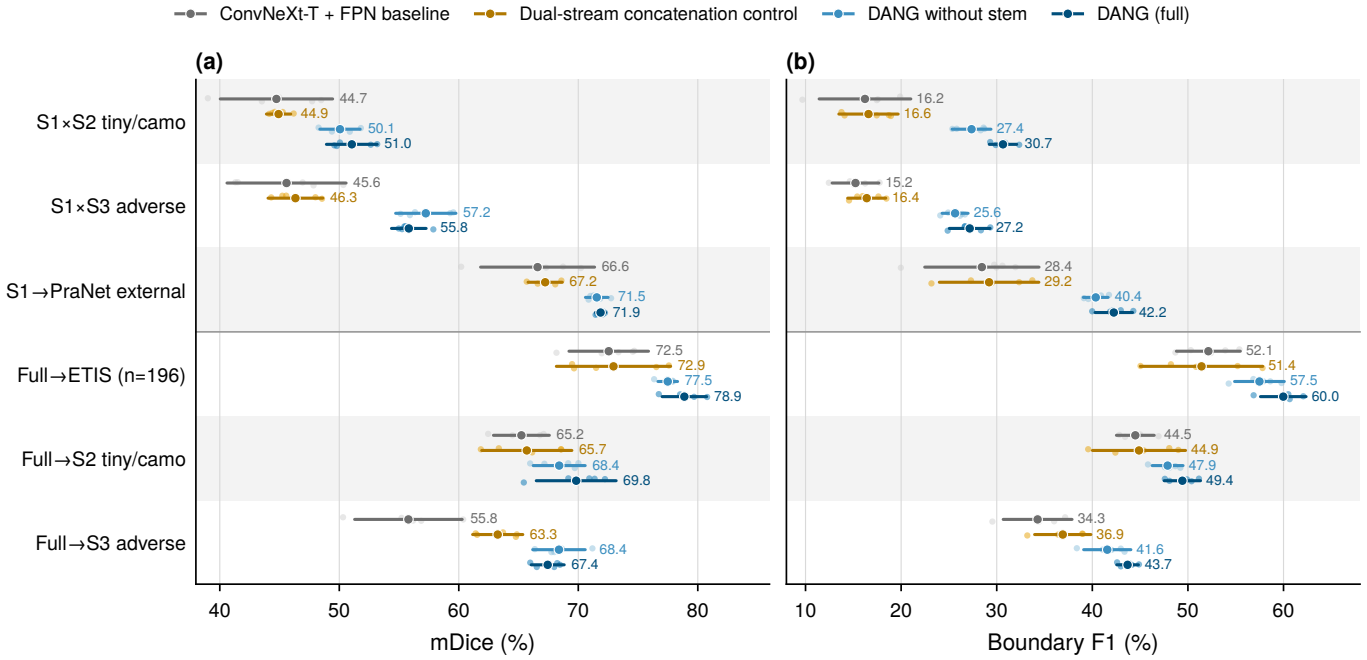


Figure 4: **The prototype/PASSE path, not the second backbone alone, carries the gain. (a)** mDice and **(b)** Boundary F1 for the four models of Table 3. Large dots and printed values are means over 5 seeds, horizontal lines are 95% t-intervals of the mean, and faint dots are individual seeds. The concatenation control (orange) stays at baseline level in the few-shot cohorts, while both models with the prototype head and PASSE (blue) separate from it. S_1 cohorts are trained on 40 Kvasir images, Full cohorts on 800.

distribution the prototype head has not seen. We keep the full model as DANG for consistency with the main experiments, but recommend the stem-free variant for deployment.

HD95. An empty prediction receives a fixed HD95 penalty of 124.5 px, which is smaller than the HD95 of 10.9% of non-empty predictions, so an empty mask can score better than a mask with a distant false positive. Empty-prediction rates are similar for DANG and the baseline (28.5% vs. 27.8% on $S_1 \times S_2$). With larger penalties the direction of the HD95 differences holds but their size shrinks, and on two cohorts they are not significant even with the default penalty. We therefore base our claims on mDice and BF1.

5.4 Efficiency

DANG is the most expensive model in Table 1: 129.9M parameters, of which 86.6M are frozen DINOv2 weights and 43.3M are trained, and 17.4 FPS against 36.5–42.4 FPS for the retrained models and 91.9 FPS for the ConvNeXt baseline. The concatenation control, which runs the same two encoders, reaches 38.4 FPS, so most of DANG's extra latency lies in the decoding path (prototype head, PASSE, strip fusion and decoder) rather than in the foundation model. Removing the stem raises throughput to 19.3 FPS. At 17 FPS DANG suits second-reader and offline review; real-time use at 25–30 Hz needs a faster GPU, an optimised implementation of the decoding path, or distillation into a smaller network (Agnihotri et al., 2025, 2026b). For reference, SAM-based polyp models are larger still (about 103M

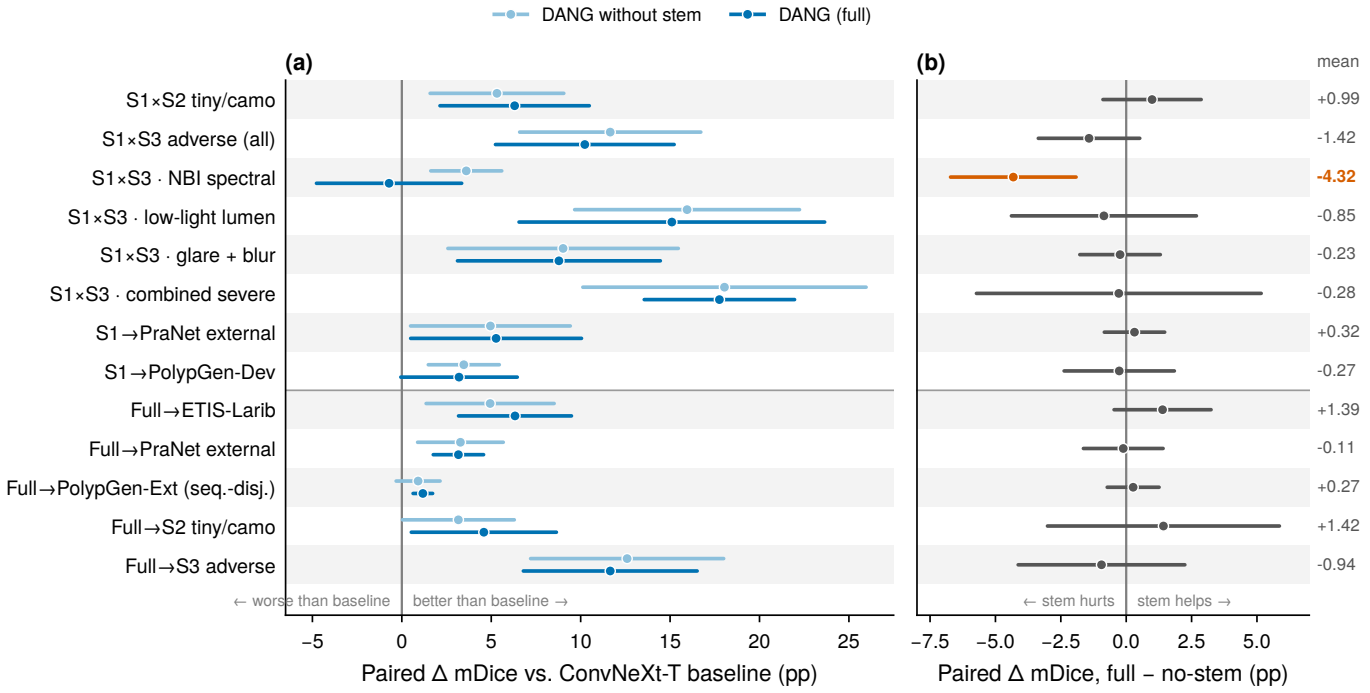


Figure 5: **Removing the Retinex stem and style calibration costs nothing, and helps on NBI. (a)** Paired mDice difference to the baseline for DANG with and without the stem. **(b)** Paired difference between the two, i.e. the contribution of the stem and style calibration. Points are means over 5 paired seeds; bars are 95% t-intervals of the paired difference. In (b) the one interval excluding zero (NBI spectral shift, -4.32 pp, $p = 0.008$) is a loss; no difference survives Holm correction. PolypGen-Ext uses the sequence-disjoint subset ($n = 900$).

parameters and 423 GFLOPs (Agnihotri et al., 2026a)).

6 Limitations

1. **Training length of the retrained baselines.** In the few-shot track, the retrained models reached their best validation score in the last 5 of 40 epochs, so they may not have converged; DANG peaked between epochs 25 and 39. Longer training could narrow the few-shot gap.

2. **Fixed few-shot subset and single source centre.** The 40-image subset is the same for all seeds, so the variance in Table 1 reflects optimisation, not the choice of images. All main models are trained on Kvasir only.

3. **Adaptive design.** PASSE thresholds and loss radii were tuned while inspecting PolypGen-Dev, S_2, S_3 and ETIS, so these cohorts are not fully held out. The extended PolypGen holdout was not used for design, and the gains there are smaller.

4. **Synthetic and derived cohorts.** S_3 is synthetic, not paired physical NBI/white-light acquisitions, and S_2 and S_3 are slices of the external cohorts.

5. **Statistical power.** Five seeds give four degrees of freedom and the standard-split experiment three seeds; several raw $p < 0.05$ effects do not survive correction.

6. **Confounded prototype loss.** Models with the prototype head also receive $\mathcal{L}_{\text{proto}}$, so their gains reflect both, and we have not run a full factorial over PASSE, the prototype loss and the strip decoder.

7. **Cost.** DANG is 2–2.5× slower than the retrained models and has 4–5× more parameters.

Ethics and data. No new patient data were collected. All experiments use public, de-identified datasets (Kvasir-SEG (Jha et al., 2020), PolypGen (Ali et al., 2023), CVC-ClinicDB, CVC-ColonDB (Tajbakhsh et al., 2016), CVC-300 and ETIS-LaribPolypDB (Silva et al., 2014)) under their research licences.

Reproducibility and AI assistance. Code, deduplication manifests, per-image evaluation logs and the scripts that regenerate every table and figure from the raw logs will be released. AI agents assisted with experiment orchestration, log aggregation, figure generation and drafting (Google Gemini; Claude Opus 5.5, Anthropic), and independent agents (Codex, OpenAI; Claude Sonnet 4.6 and Claude Opus 5.5, Anthropic) audited the aggregations, code-to-equation consistency and statistics. The authors take responsibility for all content.

7 Conclusion

DANG uses a frozen DINOv2 through a prototype margin head, a prototype-anchored support envelope and an uncertainty-guided strip decoder. When labels are scarce it outperforms PraNet, Polyp-PVT and EMCAD retrained under the same protocol, by up to 16.6 pp mDice and 20.8 pp BF1 over the strongest of them, and on the standard split it matches Polyp-PVT on four test sets and exceeds it on ETIS. A same-size concatenation control shows that the gain comes from how the foundation features are used, not from their parameters. With abundant labels DANG ties the best retrained models, and it is 2–2.5× slower. Next steps are longer few-shot training of the baselines, resampled few-shot subsets, a lighter backbone and an optimised decoding path, and evaluation on physical NBI acquisitions.

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